

Measurements

Measurements as Matrices

→ Projective Measurement

Notes on GitHub
utsavkaria-tech

$$p(a) = \text{Tr}(P_a \rho)$$

$$\text{Posterior state} \rightarrow \frac{P_a \rho P_a}{\text{Tr}(P_a \rho)}$$

Tetrahedral measurements

$$\{ |\phi_0\rangle\langle\phi_0|, |\phi_1\rangle\langle\phi_1|, |\phi_2\rangle\langle\phi_2|, |\phi_3\rangle\langle\phi_3| \}$$

↓

Form a tetrahedron on
Blochs here

$$P_a = \frac{|\phi_a\rangle\langle\phi_a|}{2} \rightarrow \text{Describes the measurement}$$

$$|\phi_0\rangle = |0\rangle$$

$$|\phi_1\rangle = \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} |1\rangle$$

$$|\phi_2\rangle = \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} e^{2\pi i/3} |1\rangle \quad |\phi_3\rangle = \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} e^{-2\pi i/3} |1\rangle$$

Measurements as Channels

→ In General ,

$$\Phi(\rho) = \sum_{a=0}^{m-1} \text{Tr}(\rho P_a) |a\rangle\langle a|$$

$$P_a = |a\rangle\langle a|$$

$$\Delta(\rho) = \langle 0|\rho|0\rangle |0\rangle\langle 0| + \langle 1|\rho|1\rangle |1\rangle\langle 1|$$

↓
Standard Basis Measurement

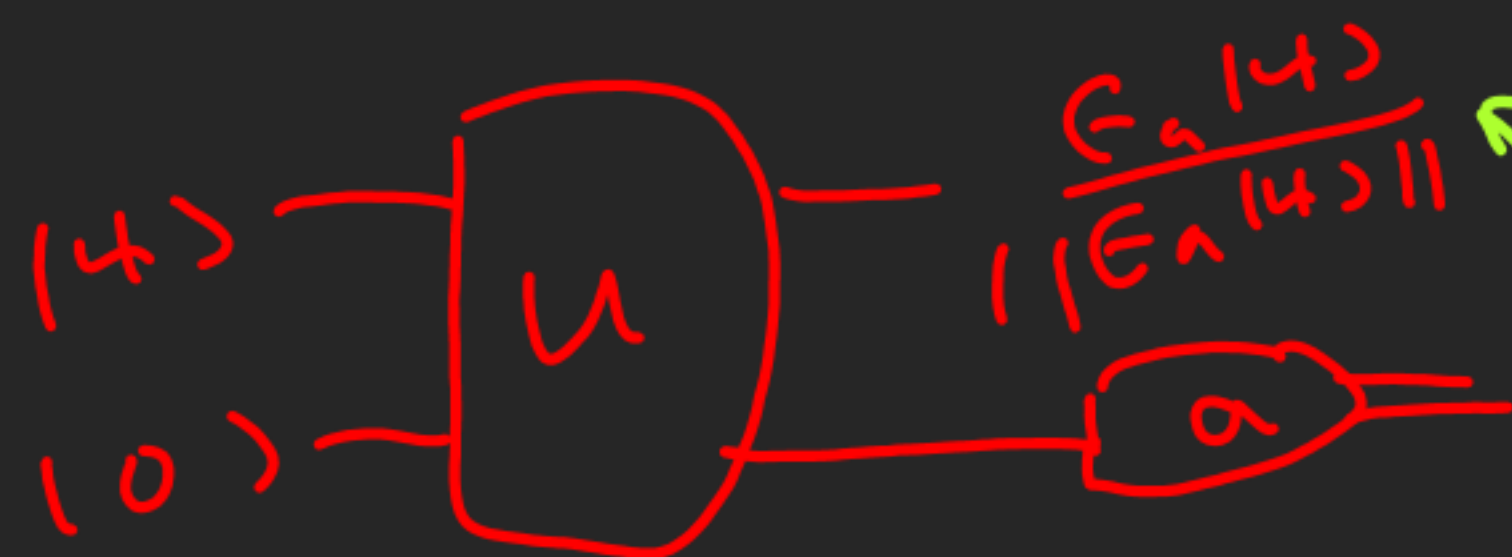
General Measurements

$$\{E_a\} \rightarrow \text{Orthogonal Projectors} \rightarrow \begin{aligned} &E_a E_b = \delta_{ab} E_a \\ &E_a = E_a^\dagger \\ &\sum_a E_a = I \end{aligned}$$

$$U = \sum_{a,b} E_a \otimes |b+a\rangle\langle a|$$

$$|\psi\rangle = |4\rangle \otimes |0\rangle$$

$$|\psi'\rangle = U|\psi\rangle = \sum_a E_a |4\rangle \otimes |a\rangle$$



$$\frac{E_a |\psi\rangle}{\|E_a |\psi\rangle\|}$$

$$p(a) = \text{Tr}(E_a \rho)$$

posterior state $\rightarrow \frac{E_a \rho E_a}{\text{Tr}(E_a \rho)}$

$$M_a = U_a \sqrt{E_a} \rightarrow |4'\rangle = U_a \left(\frac{\sqrt{E_a} |4\rangle}{\|\sqrt{E_a} |4\rangle\|} \right)$$

↓
we can freely choose POVM

Partial Measurements

$(x, z) \mapsto \rho \rightarrow \text{measuring } X \quad b, c \in \{0, \dots, N-1\}$

$$\rho = \sum_{b,c} |b\rangle\langle c| \otimes \rho_{b,c}$$

$$\phi(\mathcal{E}) = \sum_{a=0}^{n-1} \text{Tr}(\rho_a \mathcal{E}) |a\rangle\langle a|$$

$$p(a) = \text{Tr}(\rho_a \rho_x) = \text{Tr}(\rho_a \text{Tr}_z(\rho)) = \text{Tr}((\rho_a \otimes I) \rho)$$

Naimark's Theorem

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Non-Destructive Measurements

